K-Means Clustering - Explanation with Working Example

K-Means Clustering is an unsupervised machine learning algorithm used to group similar data points into clusters. The algorithm divides the dataset into K distinct clusters based on similarity. Each cluster has a centroid, which represents the center of that cluster.

How K-Means Works

1. Choose the number of clusters (K).
2. Randomly initialize K centroids.
3. Assign each data point to the nearest centroid.
4. Recalculate the centroids using the mean of assigned points.
5. Repeat steps 3 and 4 until the centroids stop changing significantly.

Distance Formula Used

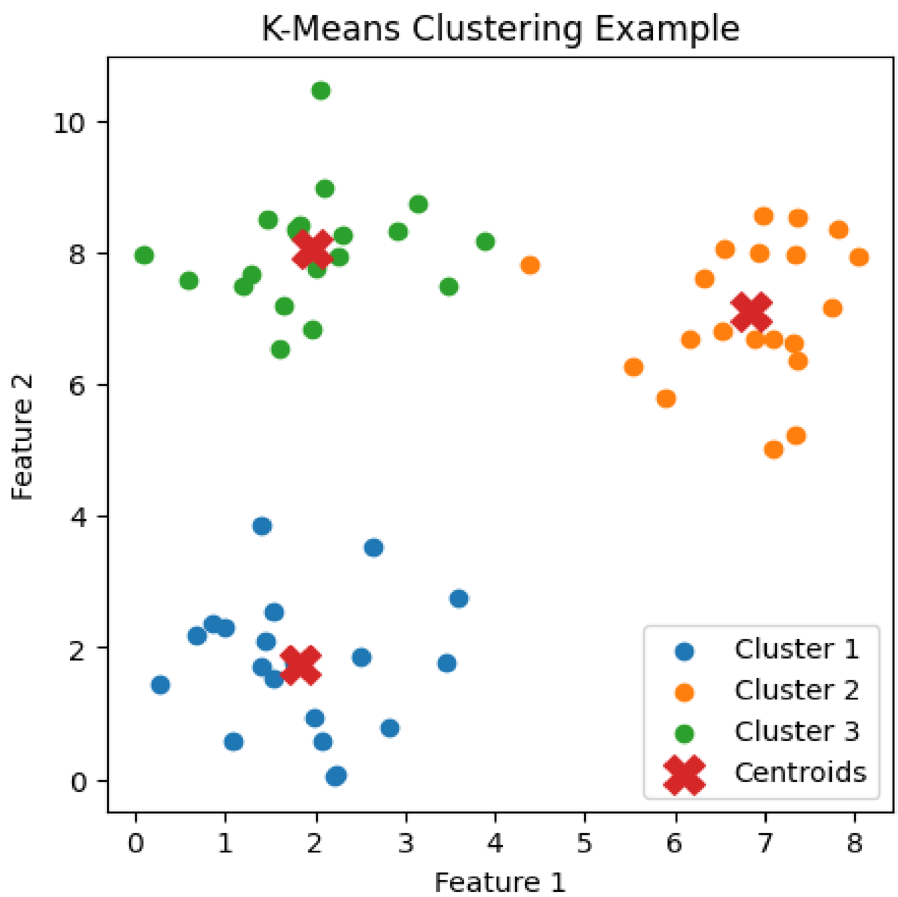
K-Means commonly uses Euclidean Distance to measure similarity between points:

$$d = \sqrt{(x_{\blacksquare} - x_{\blacksquare})^2 + (y_{\blacksquare} - y_{\blacksquare})^2}$$

Working Example

Suppose we have customer data based on spending habits and income. We want to divide customers into 3 groups using K-Means clustering.

The algorithm: Starts with 3 random centroids. Assigns each customer to the nearest centroid. Updates centroids repeatedly. Finally forms 3 meaningful customer groups.



Advantages of K-Means

Simple and easy to understand. Works efficiently on large datasets. Fast convergence. Useful for customer segmentation, image compression, and recommendation systems.

Limitations of K-Means

The value of K must be chosen manually. Sensitive to outliers. May not work well with irregularly shaped clusters.

Conclusion

K-Means clustering is one of the most popular clustering algorithms in machine learning. It helps identify hidden patterns and groups in data without labeled outputs. Its simplicity and efficiency make it widely used in real-world applications.